```
saimakapoor@LAPTOP-CT18M4UK:~$ cd ~/PES1UG24CS403_PES_VCS
saimakapoor@LAPTOP-CT18M4UK:~/PES1UG24CS403_PES_VCS$ ls
Makefile    index.c         object.o       test_objects.c    tree.c
README.md   index.h         pes.c          test_objects.o    tree.h
commit.c    object.c        pes.h          test_sequence.sh
commit.h    object.c.save   test_objects   test_tree.c
```

```
saimakapoor@LAPTOP-CT18M4UK:~/PES1UG24CS403_PES_VCS$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e
5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed022
1bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
```

```
All Phase 1 tests passed.
saimakapoor@LAPTOP-CT18M4UK:~/PES1UG24CS403_PES_VCS$ find .pes/objects -type
 f
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f4
76
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566f
cf
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc
12
```

```
saimakapoor@LAPTOP-CT18M4UK:~/PES1UG24CS403_PES_VCS$ nano tree.c
saimakapoor@LAPTOP-CT18M4UK:~/PES1UG24CS403_PES_VCS$ make clean
rm -f pes test_objects test_tree object.o tree.o index.o commit.o pes.o test
_objects.o test_tree.o
rm -rf .pes
saimakapoor@LAPTOP-CT18M4UK:~/PES1UG24CS403_PES_VCS$ make test_tree
gcc -Wall -Wextra -O2 -c test_tree.c -o test_tree.o
gcc -Wall -Wextra -O2 -c object.c -o object.o
object.c: In function ‘object_read’:
object.c:86:5: warning: ignoring return value of ‘fread’ declared with attri
bute ‘warn_unused_result’ [-Wunused-result]
   86 |     fread(buf, 1, size, f);
      |     ^~~~~~~~~~~~~~~~~~~~~~
gcc -Wall -Wextra -O2 -c tree.c -o tree.o
tree.c:5:10: fatal error: object.h: No such file or directory
    5 | #include "object.h"
      |          ^~~~~~~~~~
compilation terminated.
make: *** [Makefile:14: tree.o] Error 1
saimakapoor@LAPTOP-CT18M4UK:~/PES1UG24CS403_PES_VCS$ nano tree.c
saimakapoor@LAPTOP-CT18M4UK:~/PES1UG24CS403_PES_VCS$ ./test_tree
-bash: ./test_tree: No such file or directory
saimakapoor@LAPTOP-CT18M4UK:~/PES1UG24CS403_PES_VCS$ find .pes/objects -type
 f
find: ‘.pes/objects’: No such file or directory
saimakapoor@LAPTOP-CT18M4UK:~/PES1UG24CS403_PES_VCS$
```

```
make: *** [Makefile:14: tree.o] Error 1
saimakapoor@LAPTOP-CT18M4UK:~/PES1UG24CS403_PES_VCS$ nano tree.c
saimakapoor@LAPTOP-CT18M4UK:~/PES1UG24CS403_PES_VCS$ ./test_tree
-bash: ./test_tree: No such file or directory
saimakapoor@LAPTOP-CT18M4UK:~/PES1UG24CS403_PES_VCS$ find .pes/objects -type
 f
find: '.pes/objects': No such file or directory
saimakapoor@LAPTOP-CT18M4UK:~/PES1UG24CS403_PES_VCS$ git add tree.c
saimakapoor@LAPTOP-CT18M4UK:~/PES1UG24CS403_PES_VCS$ git commit -m "Phase 2:
 fixed tree_from_index signature and minimal implementation"
[main a08637c] Phase 2: fixed tree_from_index signature and minimal implemen
tation
 1 file changed, 63 insertions(+), 109 deletions(-)
saimakapoor@LAPTOP-CT18M4UK:~/PES1UG24CS403_PES_VCS$ git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 4 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 1.04 KiB | 1.04 MiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/SaimaKapoor/PES1UG24CS403_PES_VCS.git
   0d7a0cf..a08637c  main -> main
saimakapoor@LAPTOP-CT18M4UK:~/PES1UG24CS403_PES_VCS$
```

```
saimakapoor@LAPTOP-CT18M4UK:~/PES1UG24CS403_PES_VCS$ nano index.c
saimakapoor@LAPTOP-CT18M4UK:~/PES1UG24CS403_PES_VCS$ make clean
rm -f pes test_objects test_tree object.o tree.o index.o commit.o pes.o test
_objects.o test_tree.o
rm -rf .pes
saimakapoor@LAPTOP-CT18M4UK:~/PES1UG24CS403_PES_VCS$ make pes
gcc -Wall -Wextra -O2 -c object.c -o object.o
object.c: In function 'object_read':
object.c:86:5: warning: ignoring return value of 'fread' declared with attri
bute 'warn_unused_result' [-Wunused-result]
   86 |     fread(buf, 1, size, f);
      |     ^~~~~~~~~~~~~~~~~~~~~~
gcc -Wall -Wextra -O2 -c tree.c -o tree.o
tree.c:5:10: fatal error: object.h: No such file or directory
    5 | #include "object.h"
      |          ^~~~~~~~~~
compilation terminated.
make: *** [Makefile:14: tree.o] Error 1
saimakapoor@LAPTOP-CT18M4UK:~/PES1UG24CS403_PES_VCS$
```

```
  GNU nano 7.2                          index.c
#include "index.h"
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <sys/stat.h>
#include <time.h>

#define INDEX_PATH ".pes/index"
#define MAX_ENTRIES 1024

static void ensure_repo() {
    mkdir(".pes", 0755);
}

// ---------- LOAD ----------

int index_load(Index *idx) {
    FILE *f = fopen(INDEX_PATH, "r");
    idx->count = 0;

    if (!f) return 0; // empty index is fine

    while (!feof(f)) {
        IndexEntry *e = &idx->entries[idx->count];

        char hash_hex[65];
        if (fscanf(f, "%o %64s %ld %ld %s\n",
                   &e->mode, hash_hex, &e->mtime, &e->size, e->path) != 5) {
            break;
        }

        // convert hex → bytes
        for (int i = 0; i < 32; i++) {
            sscanf(hash_hex + i * 2, "%2hhx", &e->id.hash[i]);
        }

        idx->count++;
```

```
compilation terminated.
make: *** [Makefile:14: tree.o] Error 1
saimakapoor@LAPTOP-CT18M4UK:~/PES1UG24CS403_PES_VCS$ nano index.c
saimakapoor@LAPTOP-CT18M4UK:~/PES1UG24CS403_PES_VCS$ ./pes init
-bash: ./pes: No such file or directory
saimakapoor@LAPTOP-CT18M4UK:~/PES1UG24CS403_PES_VCS$ echo "hello" > file1.tx
t
saimakapoor@LAPTOP-CT18M4UK:~/PES1UG24CS403_PES_VCS$ echo "world" > file2.tx
t
saimakapoor@LAPTOP-CT18M4UK:~/PES1UG24CS403_PES_VCS$ ./pes add file1.txt fil
e2.txt
-bash: ./pes: No such file or directory
saimakapoor@LAPTOP-CT18M4UK:~/PES1UG24CS403_PES_VCS$ ./pes status
-bash: ./pes: No such file or directory
saimakapoor@LAPTOP-CT18M4UK:~/PES1UG24CS403_PES_VCS$ cat .pes/index
cat: .pes/index: No such file or directory
saimakapoor@LAPTOP-CT18M4UK:~/PES1UG24CS403_PES_VCS$
```

```
saimakapoor@LAPTOP-CT18M4UK:~/PES1UG24CS403_PES_VCS$ nano pes.c
saimakapoor@LAPTOP-CT18M4UK:~/PES1UG24CS403_PES_VCS$ make clean
rm -f pes test_objects test_tree object.o tree.o index.o commit.o pes.o test
_objects.o test_tree.o
rm -rf .pes
saimakapoor@LAPTOP-CT18M4UK:~/PES1UG24CS403_PES_VCS$ make pes
gcc -Wall -Wextra -O2 -c object.c -o object.o
object.c: In function ‘object_read’:
object.c:86:5: warning: ignoring return value of ‘fread’ declared with attri
bute ‘warn_unused_result’ [-Wunused-result]
   86 |     fread(buf, 1, size, f);
      |     ^~~~~~~~~~~~~~~~~~~~~~
gcc -Wall -Wextra -O2 -c tree.c -o tree.o
tree.c:5:10: fatal error: object.h: No such file or directory
    5 | #include "object.h"
      |          ^~~~~~~~~~
compilation terminated.
make: *** [Makefile:14: tree.o] Error 1
saimakapoor@LAPTOP-CT18M4UK:~/PES1UG24CS403_PES_VCS$ ./pes init
-bash: ./pes: No such file or directory
saimakapoor@LAPTOP-CT18M4UK:~/PES1UG24CS403_PES_VCS$ echo "Hello" > hello.tx
t
saimakapoor@LAPTOP-CT18M4UK:~/PES1UG24CS403_PES_VCS$ ./pes add hello.txt
-bash: ./pes: No such file or directory
saimakapoor@LAPTOP-CT18M4UK:~/PES1UG24CS403_PES_VCS$ ./pes commit -m "Initia
l commit"
-bash: ./pes: No such file or directory
saimakapoor@LAPTOP-CT18M4UK:~/PES1UG24CS403_PES_VCS$ echo "World" >> hello.t
xt
saimakapoor@LAPTOP-CT18M4UK:~/PES1UG24CS403_PES_VCS$ ./pes add hello.txt
-bash: ./pes: No such file or directory
saimakapoor@LAPTOP-CT18M4UK:~/PES1UG24CS403_PES_VCS$ ./pes commit -m "Update
 file"
-bash: ./pes: No such file or directory
saimakapoor@LAPTOP-CT18M4UK:~/PES1UG24CS403_PES_VCS$ ./pes log
-bash: ./pes: No such file or directory
saimakapoor@LAPTOP-CT18M4UK:~/PES1UG24CS403_PES_VCS$ ./pes commit -m "Initia
l commit"
-bash: ./pes: No such file or directory
saimakapoor@LAPTOP-CT18M4UK:~/PES1UG24CS403_PES_VCS$
```

```
-bash: ./pes: No such file or directory
saimakapoor@LAPTOP-CT18M4UK:~/PES1UG24CS403_PES_VCS$ find .pes -type f | sor
t
find: ‘.pes’: No such file or directory
saimakapoor@LAPTOP-CT18M4UK:~/PES1UG24CS403_PES_VCS$ git add commit.c pes.c
saimakapoor@LAPTOP-CT18M4UK:~/PES1UG24CS403_PES_VCS$ git commit -m "Phase 4:
 implemented commit and head tracking"
[main 3c4fd8e] Phase 4: implemented commit and head tracking
 2 files changed, 73 insertions(+), 190 deletions(-)
saimakapoor@LAPTOP-CT18M4UK:~/PES1UG24CS403_PES_VCS$ git push
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 4 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 1.18 KiB | 1.18 MiB/s, done.
Total 4 (delta 2), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/SaimaKapoor/PES1UG24CS403_PES_VCS.git
   81099bc..3c4fd8e  main -> main
saimakapoor@LAPTOP-CT18M4UK:~/PES1UG24CS403_PES_VCS$
```